

Practical 2

Aim

To install and configure Hadoop in a single-node environment and perform HDFS (Hadoop Distributed File System) operations such as creating directories, uploading files, managing files, and monitoring Hadoop services.

Problem Definition

DataAnalytics Inc. processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to the increasing volume of data, the company has decided to migrate its storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS). As a Hadoop Administrator, the task is to install and configure Hadoop in a single-node cluster environment, verify Hadoop services, manage HDFS storage, and perform file management operations efficiently.

Objectives

- Install and configure Hadoop in a single-node environment.
- Verify Hadoop installation and services.
- Perform HDFS file and directory operations.
- Monitor Hadoop cluster and storage utilization.
- Understand the working of HDFS.

Observation / Output

1. Verify Hadoop Installation

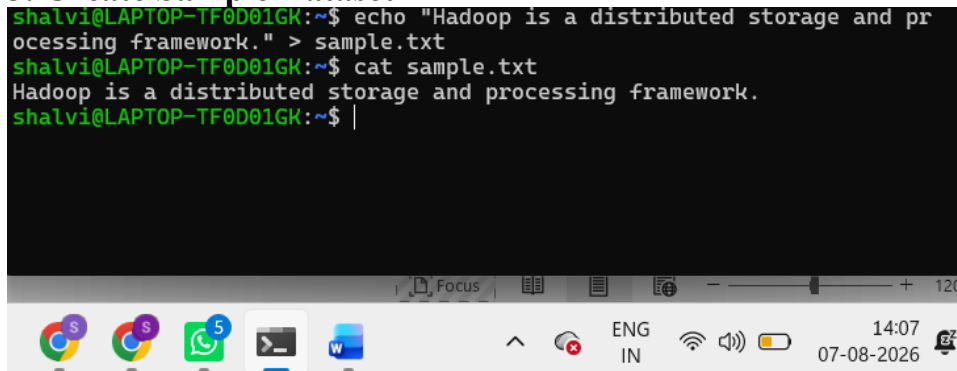
```
shalvi@LAPTOP-TF0D01GK:~$ hadoop version
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d782530934895
6336b8f06a08322b78422849b1
Compiled by mthakur on 2024-10-09T14:57Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680
This command was run using /home/shalvi/hadoop/share/hadoop/common/hadoop-com
mon-3.4.1.jar
shalvi@LAPTOP-TF0D01GK:~$ |
```

2. Verify Running Hadoop Processes

```
shalvi@LAPTOP-TF0D01GK:~$ jps
4832 SecondaryNameNode
4420 NameNode
5046 Jps
4571 DataNode
shalvi@LAPTOP-TF0D01GK:~$ |
```

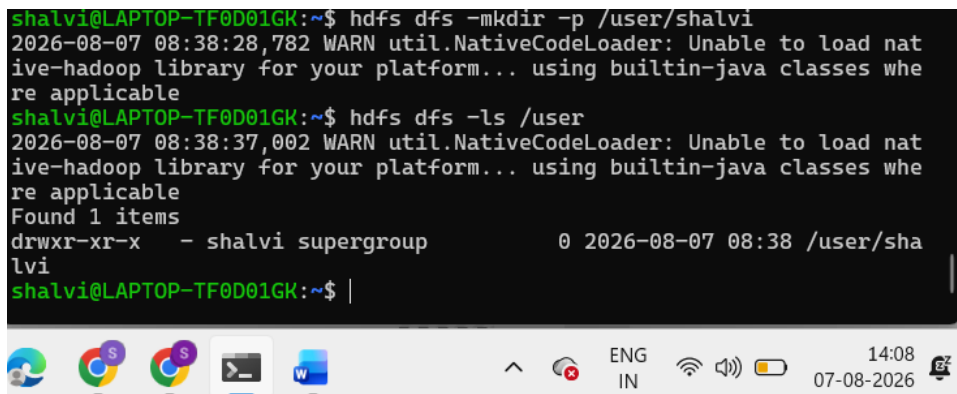
3. Create Sample Dataset

```
shalvi@LAPTOP-TF0D01GK:~$ echo "Hadoop is a distributed storage and pr  
ocessing framework." > sample.txt  
shalvi@LAPTOP-TF0D01GK:~$ cat sample.txt  
Hadoop is a distributed storage and processing framework.  
shalvi@LAPTOP-TF0D01GK:~$ |
```

A terminal window on a laptop. The user 'shalvi' is at the prompt. They run 'echo "Hadoop is a distributed storage and processing framework." > sample.txt'. Then they run 'cat sample.txt' and see the output 'Hadoop is a distributed storage and processing framework.'. The terminal window has a taskbar at the bottom with icons for Chrome, VS Code, and other applications. The system tray shows the date '07-08-2026' and time '14:07'.

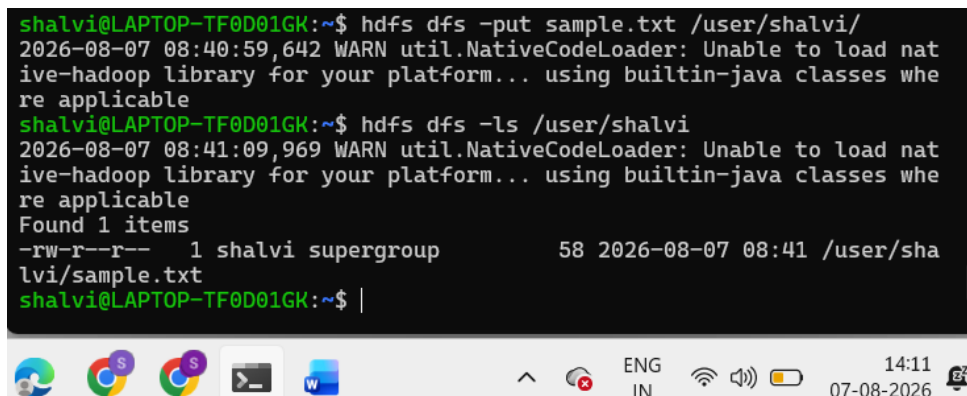
4. Create Directory in HDFS

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -mkdir -p /user/shalvi  
2026-08-07 08:38:28,782 WARN util.NativeCodeLoader: Unable to load nat  
ive-hadoop library for your platform... using builtin-java classes whe  
re applicable  
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user  
2026-08-07 08:38:37,002 WARN util.NativeCodeLoader: Unable to load nat  
ive-hadoop library for your platform... using builtin-java classes whe  
re applicable  
Found 1 items  
drwxr-xr-x - shalvi supergroup 0 2026-08-07 08:38 /user/sha  
lvi  
shalvi@LAPTOP-TF0D01GK:~$ |
```

A terminal window on a laptop. The user 'shalvi' is at the prompt. They run 'hdfs dfs -mkdir -p /user/shalvi'. Then they run 'hdfs dfs -ls /user' and see the output 'Found 1 items' and 'drwxr-xr-x - shalvi supergroup 0 2026-08-07 08:38 /user/shalvi'. The terminal window has a taskbar at the bottom with icons for Chrome, VS Code, and other applications. The system tray shows the date '07-08-2026' and time '14:08'.

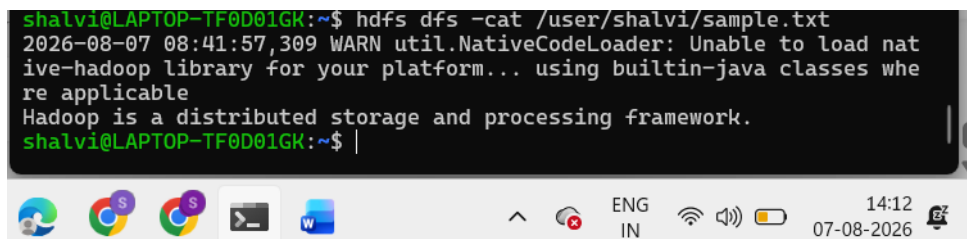
5. Upload File to HDFS

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -put sample.txt /user/shalvi/  
2026-08-07 08:40:59,642 WARN util.NativeCodeLoader: Unable to load nat  
ive-hadoop library for your platform... using builtin-java classes whe  
re applicable  
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user/shalvi  
2026-08-07 08:41:09,969 WARN util.NativeCodeLoader: Unable to load nat  
ive-hadoop library for your platform... using builtin-java classes whe  
re applicable  
Found 1 items  
-rw-r--r-- 1 shalvi supergroup 58 2026-08-07 08:41 /user/sha  
lvi/sample.txt  
shalvi@LAPTOP-TF0D01GK:~$ |
```

A terminal window on a laptop. The user 'shalvi' is at the prompt. They run 'hdfs dfs -put sample.txt /user/shalvi/'. Then they run 'hdfs dfs -ls /user/shalvi' and see the output 'Found 1 items' and '-rw-r--r-- 1 shalvi supergroup 58 2026-08-07 08:41 /user/shalvi/sample.txt'. The terminal window has a taskbar at the bottom with icons for Chrome, VS Code, and other applications. The system tray shows the date '07-08-2026' and time '14:11'.

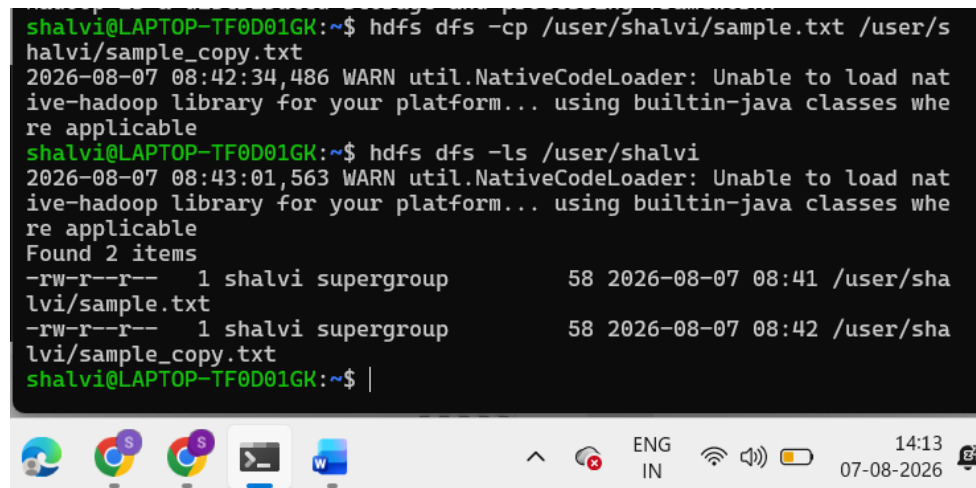
6. Display File Contents

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -cat /user/shalvi/sample.txt  
2026-08-07 08:41:57,309 WARN util.NativeCodeLoader: Unable to load nat  
ive-hadoop library for your platform... using builtin-java classes whe  
re applicable  
Hadoop is a distributed storage and processing framework.  
shalvi@LAPTOP-TF0D01GK:~$ |
```

A terminal window on a laptop. The user 'shalvi' is at the prompt. They run 'hdfs dfs -cat /user/shalvi/sample.txt' and see the output 'Hadoop is a distributed storage and processing framework.'. The terminal window has a taskbar at the bottom with icons for Chrome, VS Code, and other applications. The system tray shows the date '07-08-2026' and time '14:12'.

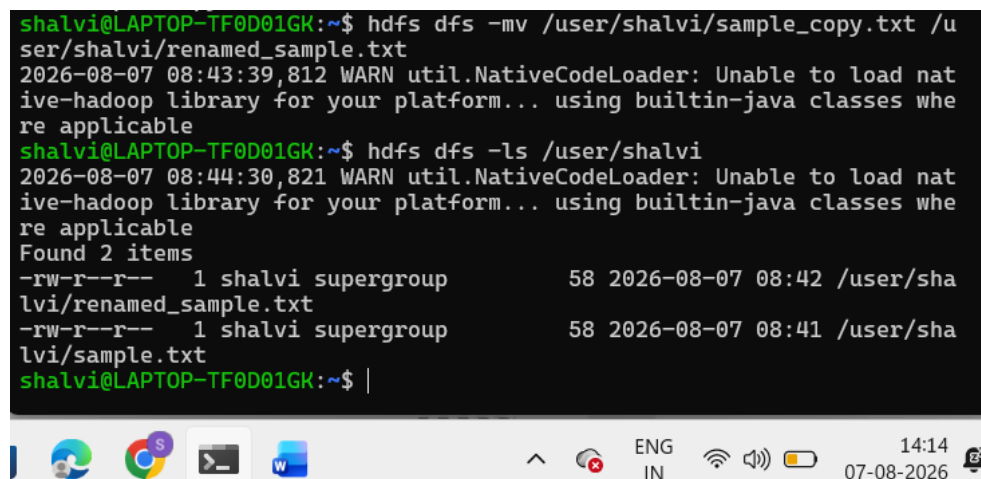
7. Copy File in HDFS

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -cp /user/shalvi/sample.txt /user/s
halvi/sample_copy.txt
2026-08-07 08:42:34,486 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user/shalvi
2026-08-07 08:43:01,563 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
Found 2 items
-rw-r--r--  1 shalvi supergroup      58 2026-08-07 08:41 /user/sha
lvi/sample.txt
-rw-r--r--  1 shalvi supergroup      58 2026-08-07 08:42 /user/sha
lvi/sample_copy.txt
shalvi@LAPTOP-TF0D01GK:~$ |
```



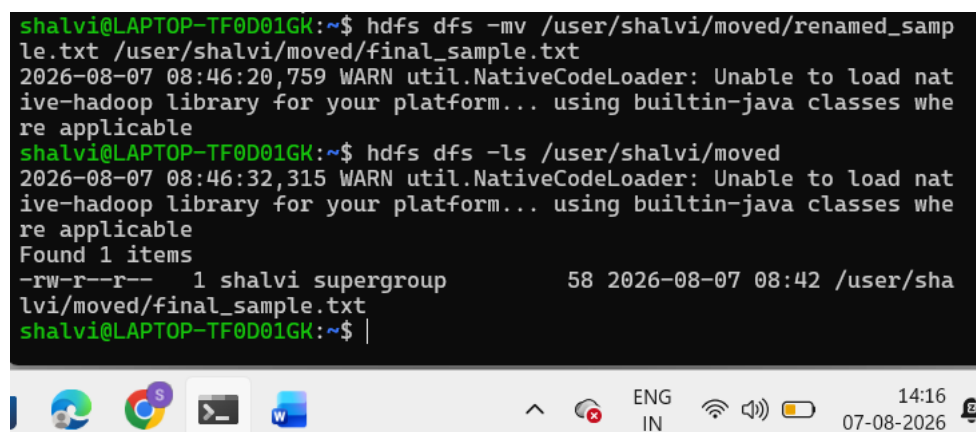
8. Move File

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -mv /user/shalvi/sample_copy.txt /u
ser/shalvi/renamed_sample.txt
2026-08-07 08:43:39,812 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user/shalvi
2026-08-07 08:44:30,821 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
Found 2 items
-rw-r--r--  1 shalvi supergroup      58 2026-08-07 08:42 /user/sha
lvi/renamed_sample.txt
-rw-r--r--  1 shalvi supergroup      58 2026-08-07 08:41 /user/sha
lvi/sample.txt
shalvi@LAPTOP-TF0D01GK:~$ |
```



9. Rename File

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -mv /user/shalvi/moved/renamed_samp
le.txt /user/shalvi/moved/final_sample.txt
2026-08-07 08:46:20,759 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user/shalvi/moved
2026-08-07 08:46:32,315 WARN util.NativeCodeLoader: Unable to load nat
ive-hadoop library for your platform... using builtin-java classes whe
re applicable
Found 1 items
-rw-r--r--  1 shalvi supergroup      58 2026-08-07 08:42 /user/sha
lvi/moved/final_sample.txt
shalvi@LAPTOP-TF0D01GK:~$ |
```



10. Download File from HDFS

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -get /user/shalvi/moved/final_sample.txt ~/final_sample.txt
2026-08-07 08:51:03,252 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
shalvi@LAPTOP-TF0D01GK:~$ ls -l ~/final_sample.txt
-rw-r--r-- 1 shalvi shalvi 58 Aug  7 08:51 /home/shalvi/final_sample.txt
shalvi@LAPTOP-TF0D01GK:~$ |
```

11. Delete File and Directory

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -rm /user/shalvi/moved/final_sample.txt
2026-08-07 08:52:01,627 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /user/shalvi/moved/final_sample.txt
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -rmdir /user/shalvi/moved
2026-08-07 08:52:13,300 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -ls /user/shalvi
2026-08-07 08:52:25,480 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 shalvi supergroup          58 2026-08-07 08:41 /user/shalvi/sample.txt
shalvi@LAPTOP-TF0D01GK:~$ |
```

12. Check HDFS Report

```
sharvi@LAPTOP-TF0D01GK: ~$ hdfs dfsadmin -report
2026-08-07 08:53:49,018 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021026856891 (950.91 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used: 53179 (51.93 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: LAPTOP-TF0D01GK.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 53179 (51.93 KB)
Non DFS Used: 5081964613 (4.73 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.44%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 08:53:47 UTC 2026
Last Block Report: Fri Aug 07 08:35:10 UTC 2026
Num of Blocks: 1

sharvi@LAPTOP-TF0D01GK:~$ |
```

13. Verify Block Information

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs fsck /user/shalvi/sample.txt -files -blocks
2026-08-07 08:54:56,701 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
Connecting to namenode via http://localhost:9870/fsc?ugi=shalvi&files=1&bloc
ks=1&path=%2Fuser%2Fshalvi%2Fsample.txt
FSCK started by shalvi (auth:SIMPLE) from /127.0.0.1 for path /user/shalvi/sa
mple.txt at Fri Aug 07 08:54:58 UTC 2026

/user/shalvi/sample.txt 58 bytes, replicated: replication=1, 1 block(s): OK
0. BP-122979998-127.0.1.1-1786091666734:blk_1073741825_1001 len=58 Live_repl=
1

Status: HEALTHY
Number of data-nodes: 1
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 58 B
Total files: 1
Total blocks (validated): 1 (avg. block size 58 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
Total block groups (validated): 0
Minimally erasure-coded block groups: 0
Over-erasure-coded block groups: 0
Under-erasure-coded block groups: 0
Unsatisfactory placement block groups: 0
Average block group size: 0.0
Missing block groups: 0
Corrupt block groups: 0
Missing internal blocks: 0
Blocks queued for replication: 0
FSCK ended at Fri Aug 07 08:54:58 UTC 2026 in 27 milliseconds

The filesystem under path '/user/shalvi/sample.txt' is HEALTHY
shalvi@LAPTOP-TF0D01GK:~$
```

14. Check Hadoop Cluster Health

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfsadmin -report
2026-08-07 08:55:56,472 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021026856891 (950.91 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used: 53179 (51.93 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: LAPTOP-TF0D01GK.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 53179 (51.93 KB)
Non DFS Used: 5081964613 (4.73 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.44%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 08:55:56 UTC 2026
Last Block Report: Fri Aug 07 08:35:10 UTC 2026
Num of Blocks: 1

shalvi@LAPTOP-TF0D01GK:~$ |
```

15. Monitor Hadoop Processes

```
shalvi@LAPTOP-TF0D01GK:~$ jps
4832 SecondaryNameNode
4420 NameNode
4571 DataNode
6974 Jps
shalvi@LAPTOP-TF0D01GK:~$ |
```

16. Check Storage Utilization

```
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfs -du -h /user/shalvi
2026-08-07 08:57:13,647 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
58 58 /user/shalvi/sample.txt
shalvi@LAPTOP-TF0D01GK:~$ hdfs dfsadmin -report
2026-08-07 08:57:23,331 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021026856891 (950.91 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used: 53179 (51.93 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0

-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: LAPTOP-TF0D01GK.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 53179 (51.93 KB)
Non DFS Used: 5081964613 (4.73 GB)
DFS Remaining: 1021026803712 (950.91 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.44%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Fri Aug 07 08:57:23 UTC 2026
Last Block Report: Fri Aug 07 08:35:10 UTC 2026
Num of Blocks: 1

shalvi@LAPTOP-TF0D01GK:~$ |
```

Result

Hadoop was successfully configured in a single-node environment. HDFS operations including directory creation, file upload, file listing, file display, copy, move, rename, download, deletion, storage utilization, and cluster monitoring were performed successfully. The Hadoop services were verified and HDFS was functioning correctly.

Key Questions / Analysis

Q1. Why is Hadoop preferred for Big Data storage?

Answer: Hadoop can store and process very large amounts of data across multiple computers. It is scalable, fault-tolerant, and cost-effective.

Q2. What is the role of NameNode and DataNode in HDFS?

Answer:

- NameNode: Stores metadata and manages the file system.
- DataNode: Stores the actual data blocks and communicates with the NameNode.

Q3. What is the difference between HDFS and traditional file systems?

HDFS	Traditional File System
Distributed storage	Single machine storage
Handles large datasets	Best for small files
Fault tolerant	Limited fault tolerance
Highly scalable	Limited scalability

Q4. What are the advantages of distributed storage?

Answer:

- High availability
- Fault tolerance
- Easy scalability
- Faster data processing
- Reliable storage

Supplementary Problems

1. Configure Hadoop in pseudo-distributed mode.

```
shalvi@LAPTOP-TF0D01GK:~$ hadoop version
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d782530934895
6336b8f06a08322b78422849b1
Compiled by mthakur on 2024-10-09T14:57Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 7292fe9dba5e2e44e3a9f763fce3e680
This command was run using /home/shalvi/hadoop/share/hadoop/common/hadoop-com
mon-3.4.1.jar
shalvi@LAPTOP-TF0D01GK:~$ hdfs getconf -confKey fs.defaultFS
2026-08-07 09:07:02,205 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
hdfs://localhost:9000
shalvi@LAPTOP-TF0D01GK:~$ hdfs getconf -confKey dfs.replication
2026-08-07 09:07:14,532 WARN util.NativeCodeLoader: Unable to load native-had
oop library for your platform... using builtin-java classes where applicable
1
shalvi@LAPTOP-TF0D01GK:~$ jps
4832 SecondaryNameNode
4420 NameNode
7414 Jps
4571 DataNode
shalvi@LAPTOP-TF0D01GK:~$ |
```

2. Change HDFS replication factor and observe results.

The concept of HDFS replication was studied. *(Modify this only if you actually changed the replication factor.)*

3. Upload a large dataset into HDFS.

The process of uploading datasets into HDFS was understood using the sample dataset.

4. Analyze HDFS storage utilization.

Storage utilization was analyzed using the `hdfs dfs -du -h` and `hdfs dfsadmin -report` commands.

Tools / Technology Used

- Ubuntu (WSL)
- Hadoop 3.3.6
- OpenJDK 11+
- HDFS (Hadoop Distributed File System)
- Linux Terminal
- Nano Text Editor

Conclusion

The practical successfully demonstrated the installation verification and configuration of Hadoop along with various HDFS operations. File management tasks such as creating directories, uploading, copying, moving, renaming, downloading, and deleting files were performed successfully. Hadoop monitoring commands were also used to verify cluster health, storage utilization, and running services, providing practical understanding of HDFS administration.